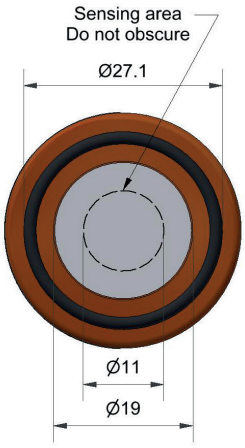
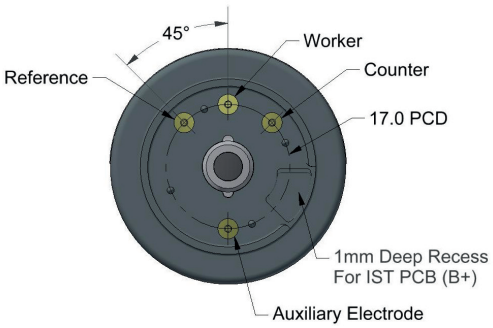
NO2-B43F/NO2-B43F+ Nitrogen Dioxide Sensor

The NO2-B43F sensor is a PPB sensor that is designed for environmental air quality applications with best-in-class baseline stability. This product is available in our standard format (NO2-B43F) and with our patented Integrated Smart Technology (NO2-B43F+) that has an IST board with a memory chip and temperature sensor integrated in the sensor. The + sensors store specific calibration, specification, and identification data on every sensor allowing plug and play operation. The on-board temperature sensor improves the accuracy and simplicity of temperature compensation algorithms.

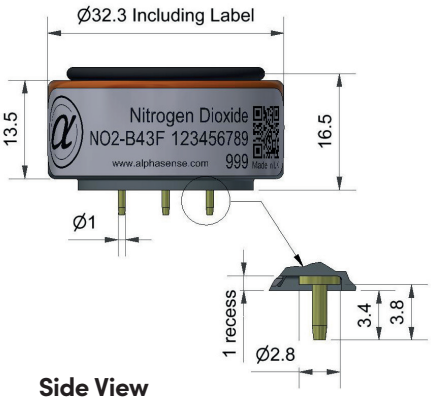
NO2-B43F Nitrogen Dioxide Sensor – 4-Electrode



Top View

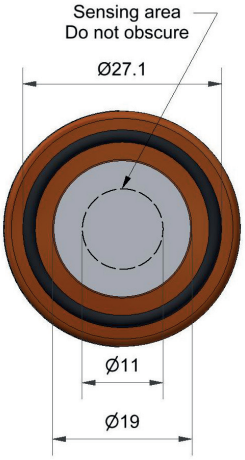


Bottom View

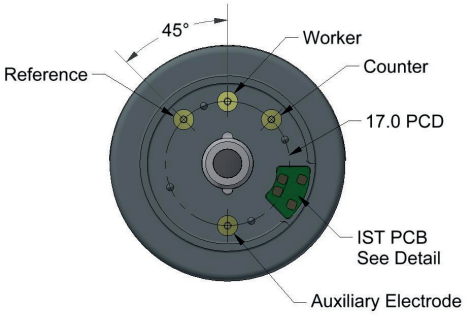


Side View

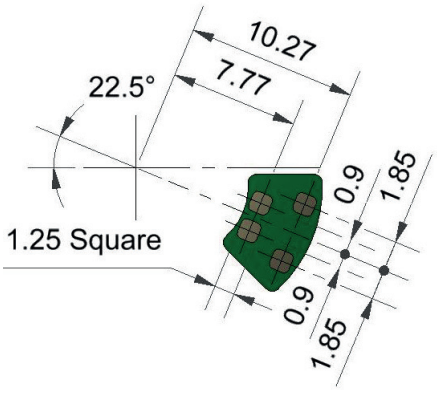
NO2-B43F+ Nitrogen Dioxide Sensor – 4-Electrode (with Integrated Smart Technology)



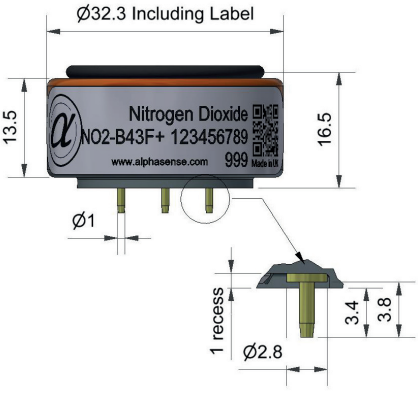
Top View



Bottom View



IST Board



Side View

B+ SERIES

INTEGRATED SMART TECHNOLOGY

Dimensions are in millimetres (± 0.15 mm).

T: +44 (0)1376 556700 **E:** sensors.alphasense@ametek.com **W:** www.alphasense.com

For further information on the performance of this sensor, on other sensors in the range or any other subject, please contact Alphasense Ltd. or visit our website at "www.alphasense.com".

Sensor Data

Performance	Sensitivity	nA/ppm at 2ppm NO_2	-200 to -650
	Response time	t90 (s) from zero to 2ppm NO_2	< 80
	Zero current	nA in zero air at 20°C	-80 to +80
	Noise*	±2 standard deviations (ppb equivalent)	15
	Range	ppm NO_2 limit of performance warranty	20
	Linearity	ppb error at full scale, linear at zero and 20ppm	< ± 0.5
	Overgas limit	NO_2 maximum ppm for stable response to gas pulse	50
	***Tested with Alphasense ISB low noise circuit**		
Lifetime	Zero drift	ppb equivalent change/year in lab air	0 to 20
	Sensitivity drift	% change/year in lab air, monthly test	-20 to -40
	Operating life	months until 50% original signal (24-month warranted)	> 24
Environmental	Sensitivity @ -20°C	% (output @ -20°C/output @ 20°C) @ 2ppm NO_2	60 to 80
	Sensitivity @ 40°C	% (output @ 50°C/output @ 20°C) @ 2ppm NO_2	95 to 115
	Zero @ -20°C	nA	0 to 25
	Zero @ 40°C	nA	-10 to 50

Cross-sensitivity	O_3	filter capacity (ppm hrs)	@ 0.5ppm	O_3	< 500
	H_2S	sensitivity	% measured gas @ 5ppm	H_2S	< -80
	NO	sensitivity	% measured gas @ 5ppm	NO	< 5
	Cl_2	sensitivity	% measured gas @ 5ppm	Cl_2	< 100
	SO_2	sensitivity	% measured gas @ 5ppm	SO_2	< -3
	CO	sensitivity	% measured gas @ 5ppm	CO	< -3
	H_2	sensitivity	% measured gas @ 100ppm	H_2	< 0.1
	C_2H_4	sensitivity	% measured gas @ 100ppm	C_2H_4	< 0.1
	NH_3	sensitivity	% measured gas @ 20ppm	NH_3	< 0.1
	CO_2	sensitivity	% measured gas @ 5% volume	CO_2	< 0.1
	Halothane	sensitivity	% measured gas @ 100ppm	Halothane	nd

Key Specifications	Temperature range	°C	-30 to 40
	Pressure range	kPa	80 to 120
	Humidity range	% rh continuous	15 to 85
	Storage period	months @ 3 to 20°C (stored in sealed pot)	6
	Load resistor	Ω (ISB circuit is recommended)	33 to 100
	Weight	g	< 13

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Figure 1 Sensitivity Temperature Dependence

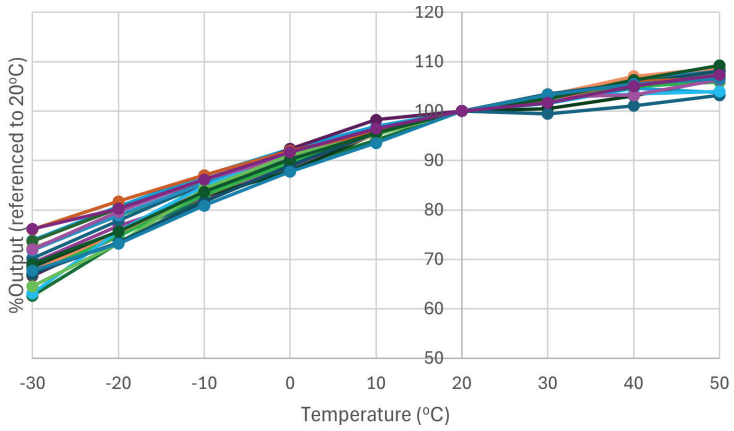


Figure 1 shows the temperature dependence of sensitivity at 2ppm NO_2.

This data is taken from a typical batch of sensors.

Figure 2 Zero Temperature Dependence

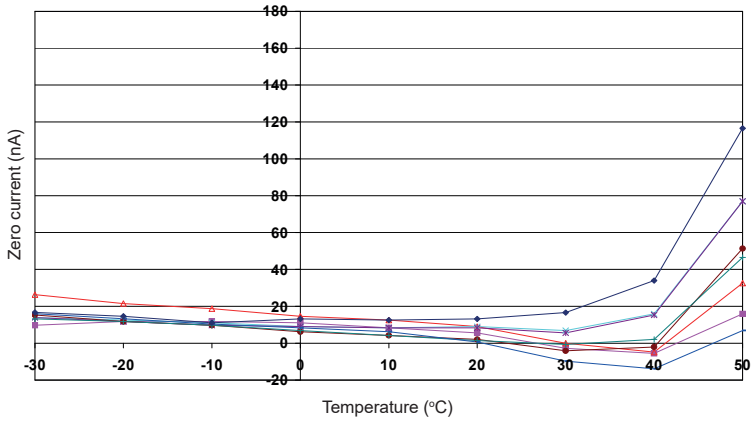
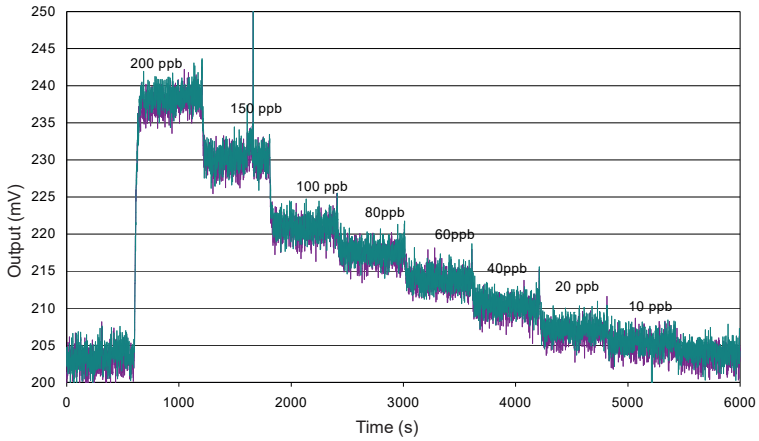


Figure 2 shows the variation in zero output of the working electrode caused by changes in temperature, expressed as nA.

This data is taken from a typical batch of sensors.

Contact Alphasense for further information on zero current correction.

Figure 3 Response from 200 ppb NO_2



With a 33 Ω load resistor, the NO2-B43F shows excellent resolution, even at the ppb level: ideal for outdoor air environmental testing.

Use of Alphasense ISB circuit reduces noise to 15ppb, with the opportunity of digital smoothing to reduce noise even further.

Offset voltage is due to intentional ISB circuit electronic offset.

IST Board Data

Interface	Communication Bus	Compatible with the 400 kHz I²C protocol
	Max. Bus Speed	Up to 1 MHz
	Input Logic Levels	High (Recessive) < 2.3 V | Low (Dominant) < 0.2 V
	Absolute Max. Input Signal	3.6 V
Electrical	Supply Voltage Range	1.7 V to 3.6 V
	Supply current – Stand-By	< 5 µA
	Supply current – Operating	< 0.15 mA (temperature reading only) < 2.15 mA (temperature reading + memory reading/writing)
	Power Supply Conditioning	Built-In 100 nF decoupling capacitor
	ESD Protection	4 kV (human body model) - Enhanced ESD / Latch-Up protection
	Bus Pins Input Capacitance	15 pF max.
Performance	Operational Temperature	–40 °C to +85 °C
	Temperature Sensor Accuracy	±1°C (-0°C to +70°C)
	Memory Data Retention	> 200 years
	Memory Write Cycles	> 4,000,000
Data & Communication	Memory IC & I2C Address	M24128X-FCU | Device Address: R - 0xA0 / W – 0xA1
	Temperature IC & I2C Address	MAX31875R0TZS+T | Device Address: R - 0x90 / W – 0x91
	Product Data Start Address	0x0900
	Calibration Data Start Address	0x0B00
	User Data Area	0x0D00 – 0x18FF (3,072 Bytes)
	CRC Polynomial	0x 01 04C1 1DB7
	Digital Signature Algorithm	SHA-256

Factory-populated data

Product Data
- Data Format Version
- Customer (OEM) ID
- Product ID
- Type of Sensor / Target Gas
- Sensor Serial Number
- End of Storage Period Date
- Sensor Replacement Date
- Product Data Checksum
- Alphasense Digital Signature
- Customer Digital Signature

Calibration
- Calibration Data Units
- Zero (clean dry air) Output
- Calibration Span
- Calibration Output
- Sensitivity
- Calibration Date
- Calibration Data Checksum
- Calibration Data Signature

Sensor Specification
- Over-gas limit
- Concentration Range
- Temperature Range Low
- Temperature Range High
- Humidity Range Low
- Humidity Range High
- Pressure Range Low
- Pressure Range High
- Specification Checksum

15,000+ locations

Customer Specific
- Custom Parameters
- Re-Calibration Due Date
- Operational Limits:
- Low | High | STEL | TWA
- Next Bump Test Due Date
- User Data Area

At the end of the product's life, do not dispose of any electronic sensor, component or instrument in the domestic waste, but contact the instrument manufacturer, Alphasense or its distributor for disposal instructions. NOTE: all sensors are tested at ambient environmental conditions unless otherwise stated. As applications of use are outside our control, the information provided is given without legal responsibility. Customers should test under their own conditions, to ensure that the sensors are suitable for their own requirements.

In the interest of continued product improvement, we reserve the right to change design features and specifications without prior notification. The data contained in this document is for guidance only. Alphasense Ltd accepts no liability for any consequential losses, injury or damage resulting from the use of this document or the information contained within.(©ALPHASENSE LTD) Doc. Ref. NO2-B43F/AUG24

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